12th International ABINIT Developer Workshop

Cubic scaling spinor GW in real space and imaginary time

Hsiao-Yi Tsai

Dr. Matteo Giantomassi, Prof. Xavier Gonze, Prof. Samuel Poncé

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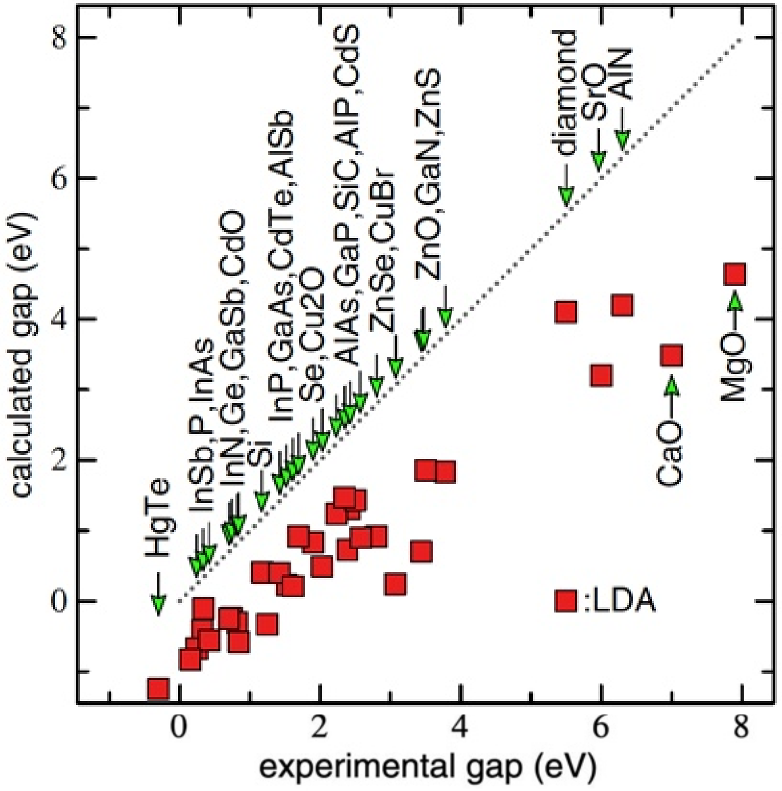
First-principles calculations help explore materials



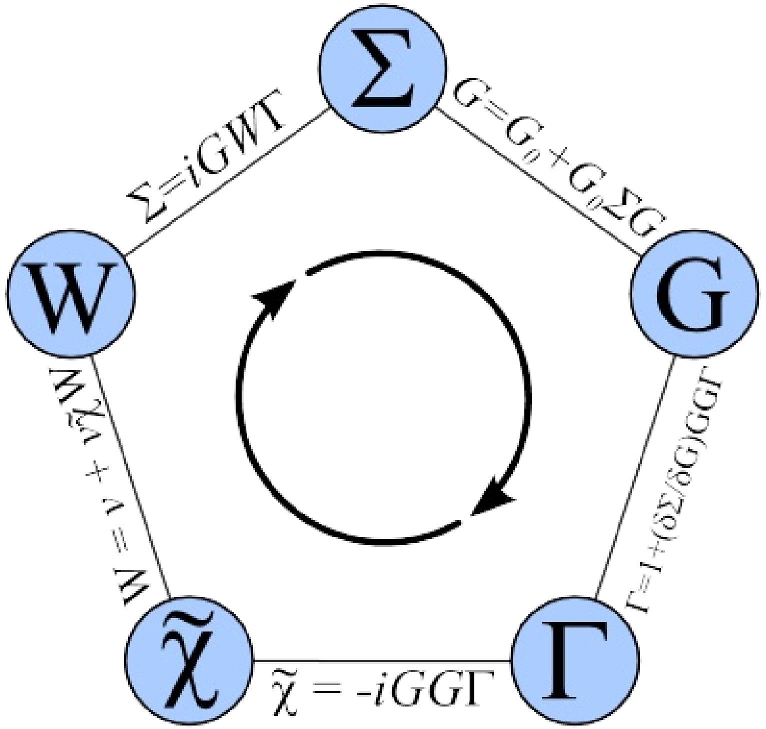
Density Functional Theory (DFT) is a cheap ab-initio method to describe ground state properties.

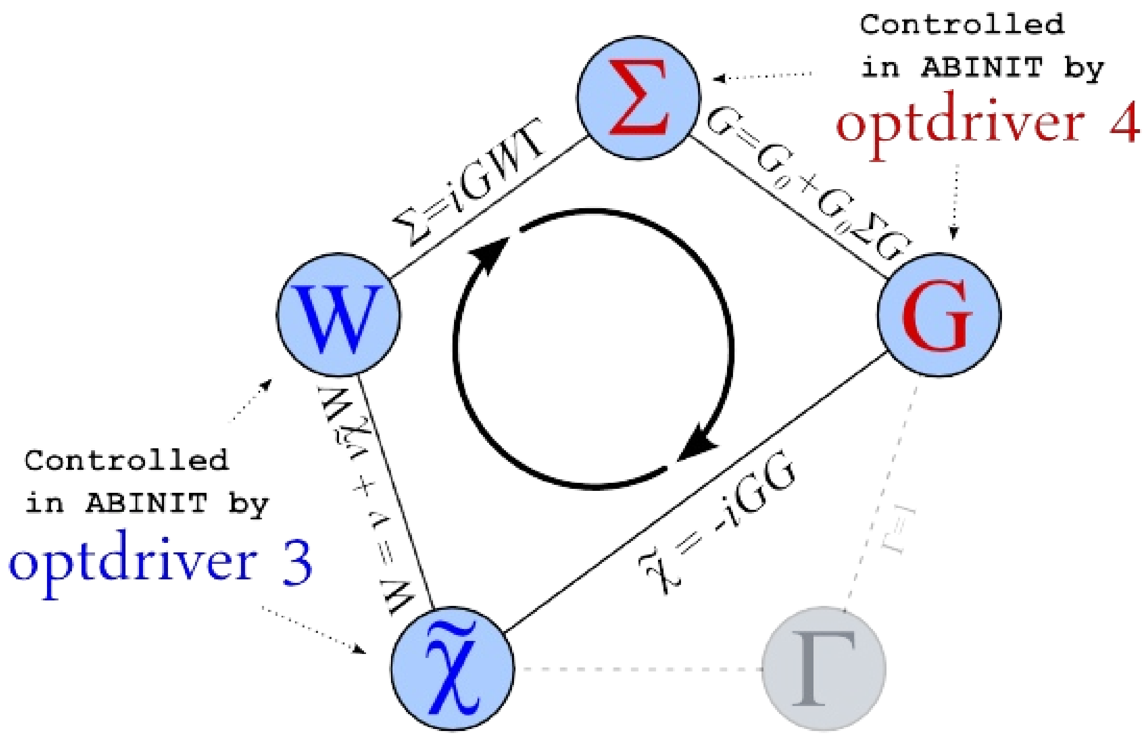
But underestimates band gaps.

M. van Schilfgaarde, T. Kotani, and S. Faleev, Phys. Rev. Lett. 96, 226402 (2006).

GW method for accurate electronic band structure

GW: A many-body perturbation method used to calculate self-energies

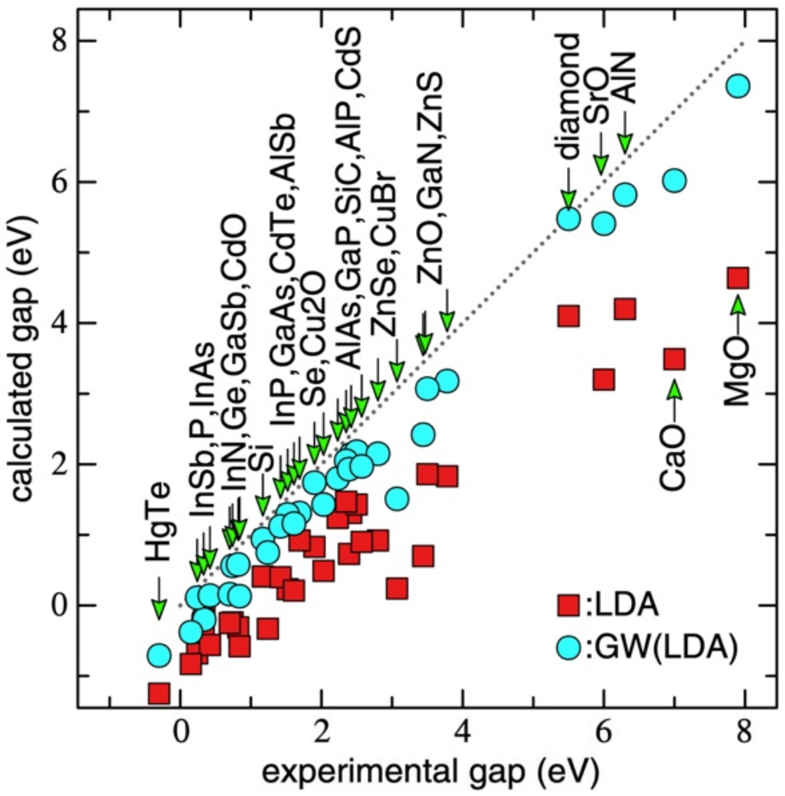


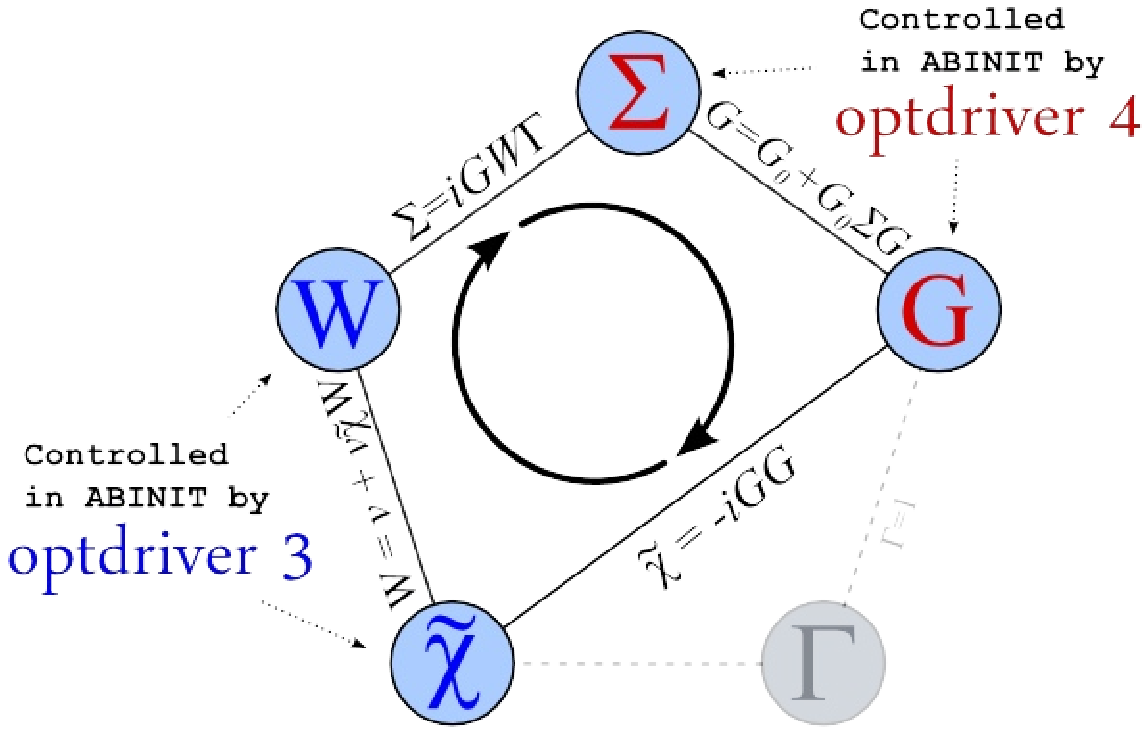


X. Gonze, B. Amadon, and et.al., Computer Physics Communications 180, 2582 (2009).
L. Hedin, Phys. Rev. 139, A796 (1965).

GW method for accurate electronic band structure

GW: A many-body perturbation method used to calculate self-energies





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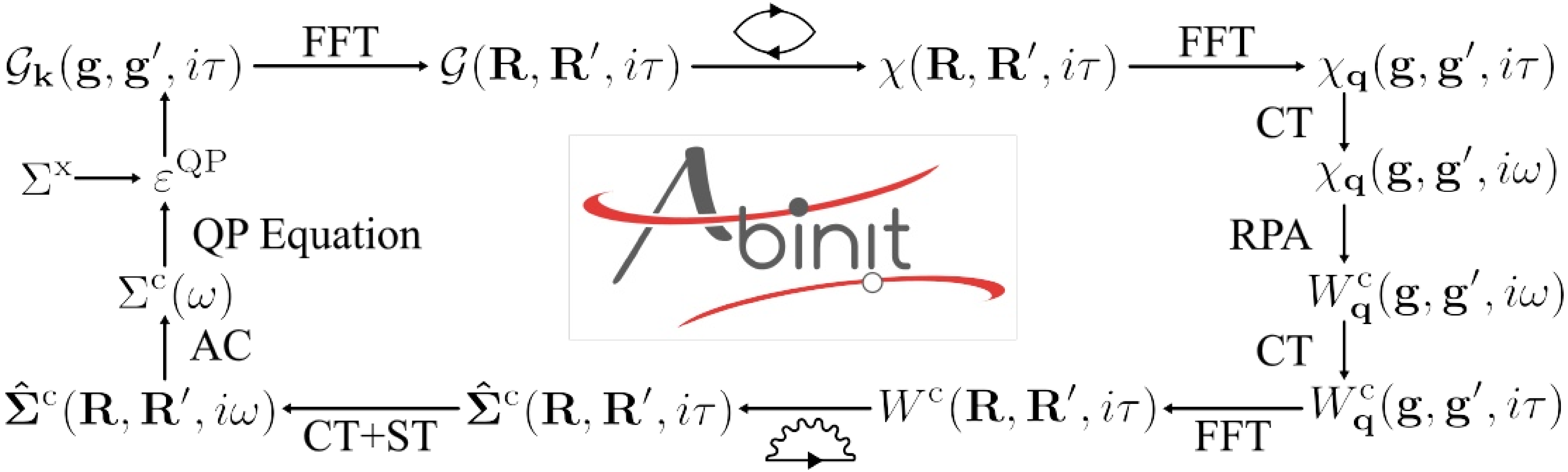
Cubic scaling GW in real-space and imaginary-time



Quartic-scaling GW

$$\Sigma(\mathbf{k},\omega) = G(\mathbf{k},\omega) * W(\mathbf{k},\omega)$$

Convolution

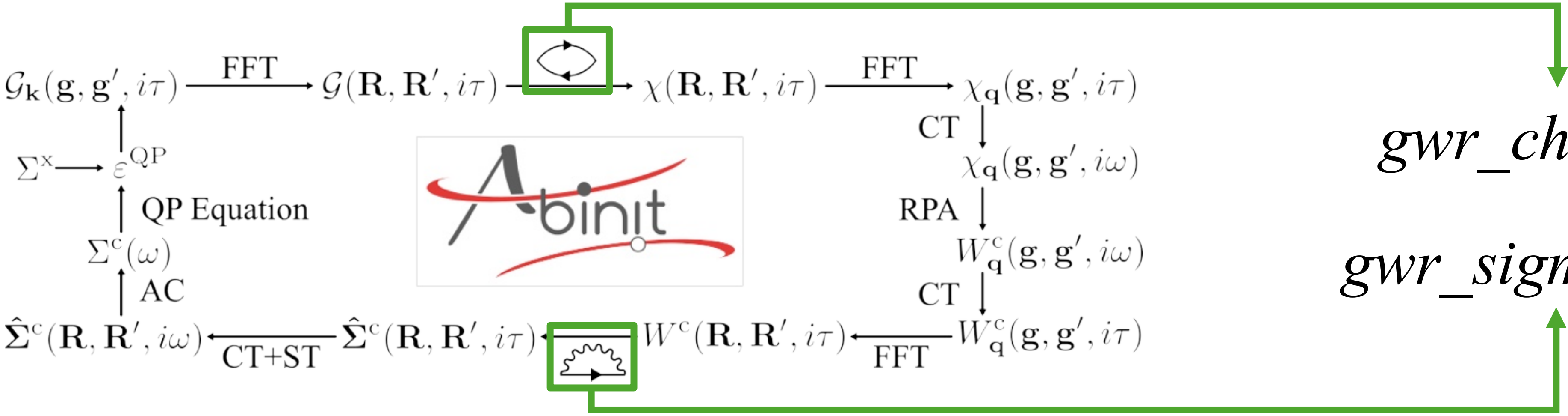
optdriver 6 Cubic-scaling GW (GWR)

$$\Sigma(\mathbf{R},i\tau) = G(\mathbf{R},i\tau) \times W(\mathbf{R},i\tau)$$

Multiplication

P. Liu, M. Kaltak, J. Klimeš, and G. Kresse, Phys. Rev. B 94

Flexibility in balancing memory and time



gwr_chi_algo

gwr_sigma_algo

1 Supercell method: 

Linear scaling wrt number of k-points

Very fast but memory demanding !!!

$$\hat{\Sigma}^c(\mathbf{R},\mathbf{R}',i\tau) \propto \underline{G(\mathbf{R},\mathbf{R}',i\tau)}W^c(\mathbf{R},\mathbf{R}',i\tau)$$

in supercell

2 Convolution method: 

Quadratic scaling wrt number of k-points

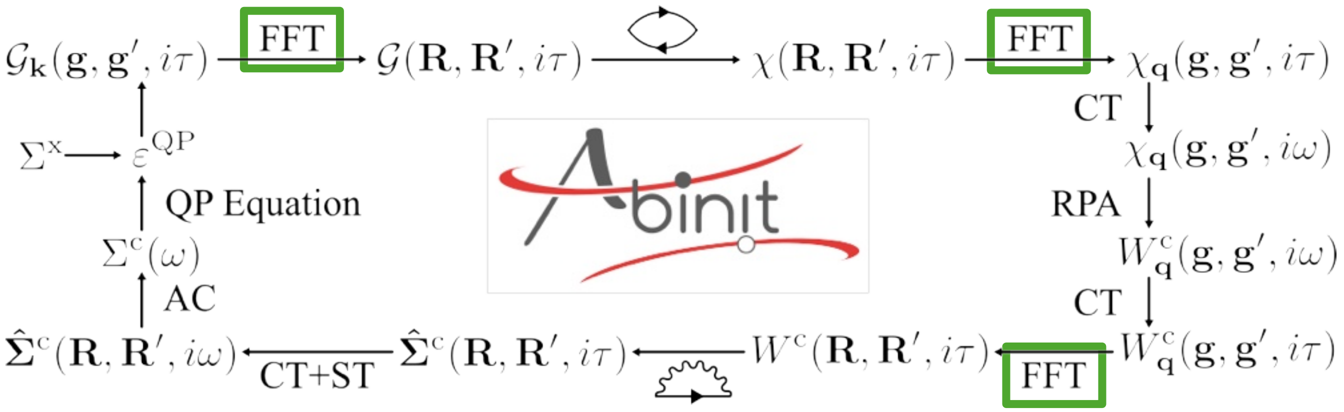
Slower but memory saving !!!

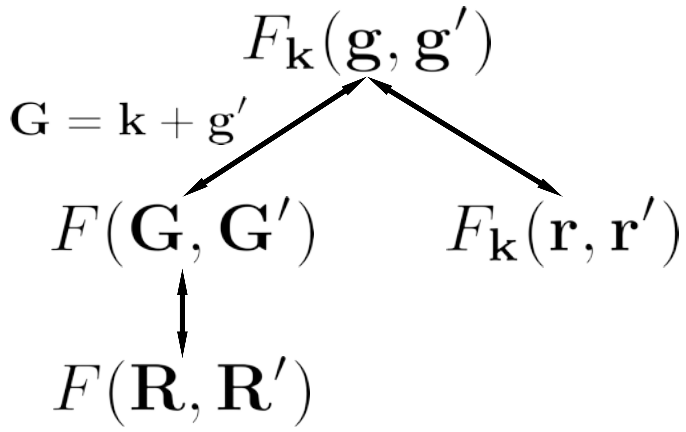
$$\hat{\Sigma}^c_{\mathbf{k}}(\mathbf{r},\mathbf{r}',i\tau) \propto \sum_{\mathbf{q}}^{\mathrm{BZ}} \underline{G_{\mathbf{k}+\mathbf{q}}(\mathbf{r},\mathbf{r}',i\tau)W^c_{\mathbf{q}}(\mathbf{r},\mathbf{r}',i\tau)}$$

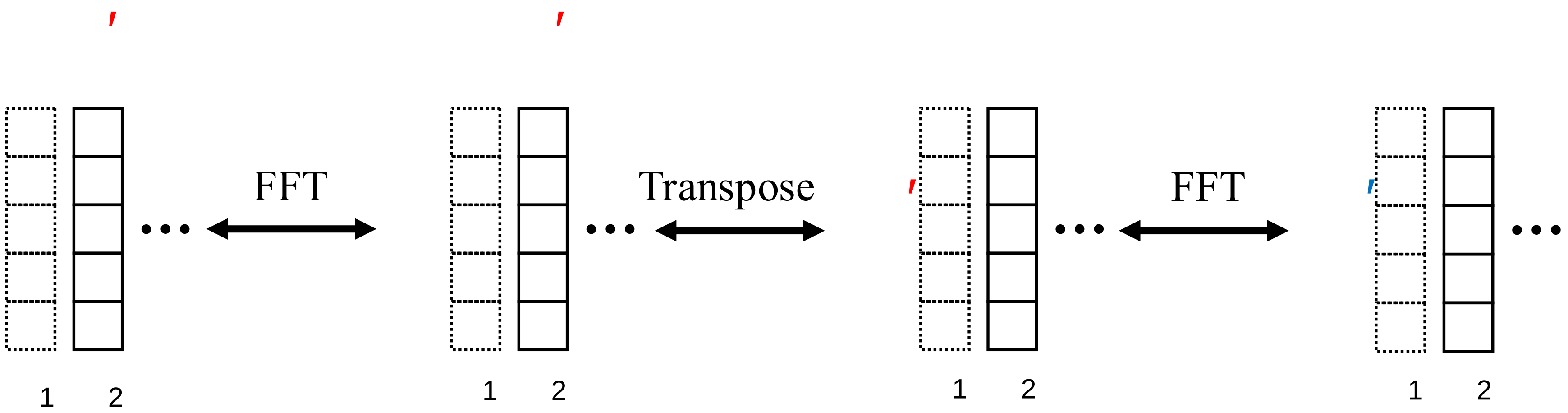
in unit-cell

Symmetrized to IBZ

Spatial Fast Fourier Transformation

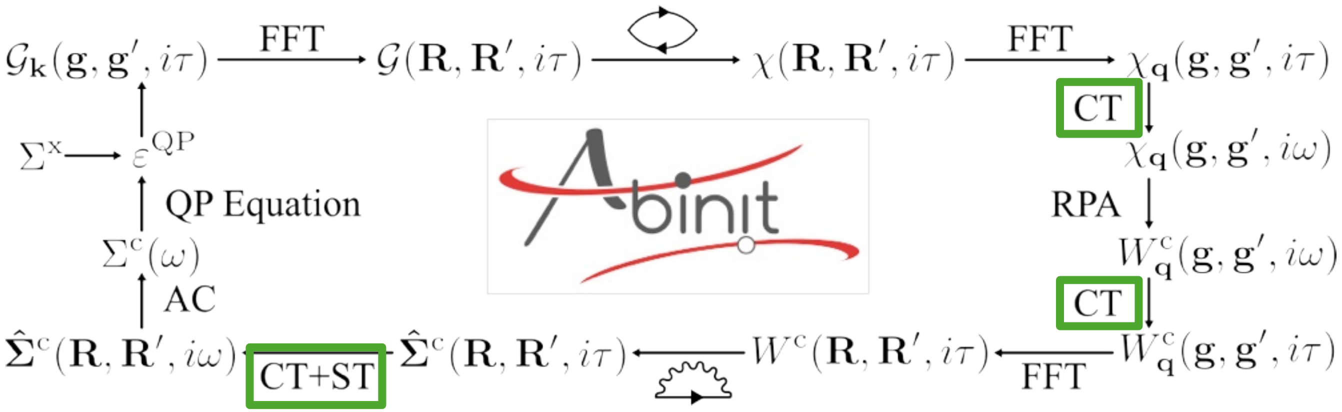






Ongoing: moving to GPU

Cubic scaling GW in real-space and imaginary-time



The selection of the grid depends on:

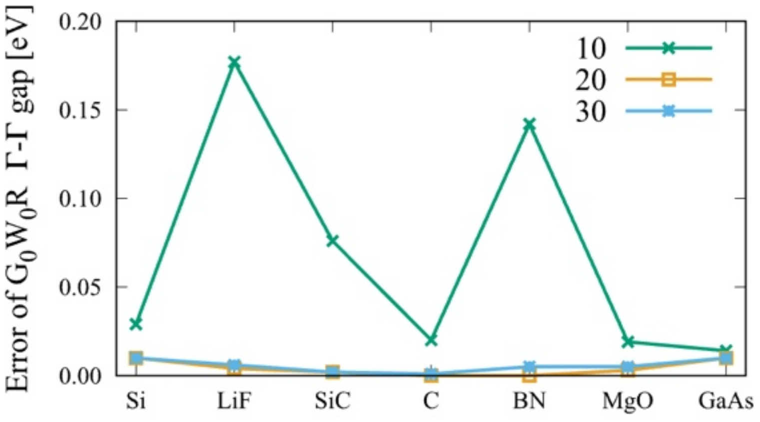
$$\text{Number of points} \quad \frac{\text{Maximum transition energy (controlled by } \textit{nband}\text{)}}{\text{Minimum transition energy (usually fundamental gap)}}$$

!!! does not support metallic systems for now !!!

CT/ST: cos/sin transformation

$$\chi(i\omega_k) = \sum_{j=1}^{N} \gamma_{kj} \cos(\omega_k \tau_j)\chi(i\tau_j)$$

GREENX provides time/frequency grids and weights



Error with respect to the conventional GW

M. Azizi, J. Wilhelm, D. Golze, F. A. Delesma, R. L. Panadés-Barrueta, P. Rinke, M. Giantomassi, and X. Gonze, Phys. Rev. B 109, 245101 (2024).

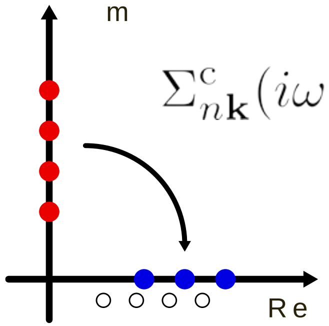
From self-energy to band structure

$$\begin{tikzcd}
 & \mathcal{G}_{\mathbf{k}}(\mathbf{g},\mathbf{g}',i\tau) \arrow[r, "\mathrm{FFT}"] & \mathcal{G}(\mathbf{R},\mathbf{R}',i\tau) \arrow[r] & \chi(\mathbf{R},\mathbf{R}',i\tau) \arrow[r, "\mathrm{FFT}"] & \chi_{\mathbf{q}}(\mathbf{g},\mathbf{g}',i\tau) \arrow[d, "\mathrm{CT}"'] \\
\Sigma^{\mathrm{x}} \arrow[r] & \varepsilon^{\mathrm{QP}} \arrow[u] & & & \chi_{\mathbf{q}}(\mathbf{g},\mathbf{g}',i\omega) \arrow[d, "\mathrm{RPA}"'] \\
 & \Sigma^{\mathrm{c}}(\omega) \arrow[u, "\boxed{\text{QP Equation}}"'] & & & W^{\mathrm{c}}_{\mathbf{q}}(\mathbf{g},\mathbf{g}',i\omega) \arrow[d, "\mathrm{CT}"'] \\
 & \hat{\Sigma}^{\mathrm{c}}(\mathbf{R},\mathbf{R}',i\omega) \arrow[u, "\boxed{\text{AC}}"'] & \hat{\Sigma}^{\mathrm{c}}(\mathbf{R},\mathbf{R}',i\tau) \arrow[l, "\mathrm{CT+ST}"] & W^{\mathrm{c}}(\mathbf{R},\mathbf{R}',i\tau) \arrow[l] & W^{\mathrm{c}}_{\mathbf{q}}(\mathbf{g},\mathbf{g}',i\tau) \arrow[l, "\mathrm{FFT}"]
\end{tikzcd}$$

Shankland-Koelling-Wood interpolation

gwr.interpolate(
ks_ebands_kpath=...,
...)

AC: Analytic Continuation



$$\Sigma^{\mathrm{c}}_{n\mathbf{k}}(i\omega) \xrightarrow{\text{Padé}} \Sigma^{\mathrm{c}}_{n\mathbf{k}}(\omega)$$

Perturbative approach

$$\varepsilon^{QP}_{n\mathbf{k}} = \varepsilon^{KS}_{n\mathbf{k}} + Z\langle\phi^{KS}_{n\mathbf{k}}|\hat{\Sigma}(\varepsilon^{KS}_{n\mathbf{k}}) - \hat{V}^{xc}|\phi^{KS}_{n\mathbf{k}}\rangle$$

$$Z \equiv \left[1 - \langle\phi^{KS}_{n\mathbf{k}}|\frac{\partial\hat{\Sigma}(\varepsilon^{KS}_{n\mathbf{k}})}{\partial\omega}|\phi^{KS}_{n\mathbf{k}}\rangle\right]^{-1}$$

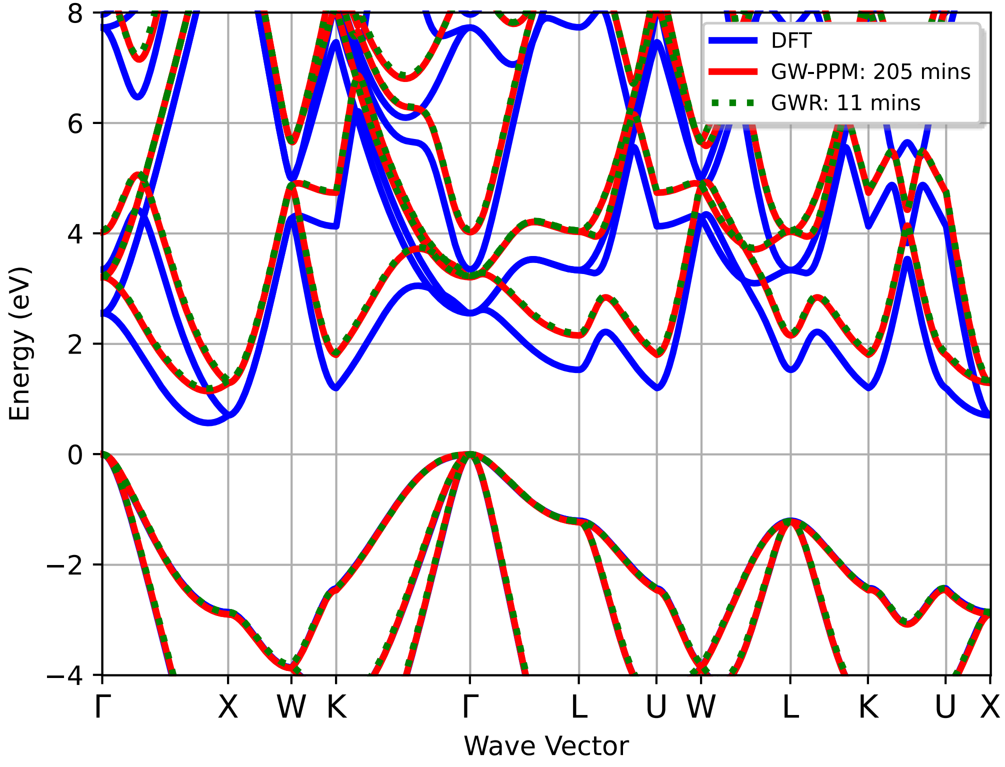
Solving QP equation

$$\varepsilon^{QP}_{n\mathbf{k}} = \varepsilon^{KS}_{n\mathbf{k}} + \langle\phi^{KS}_{n\mathbf{k}}|\hat{\Sigma}(\varepsilon^{QP}_{n\mathbf{k}}) - \hat{V}^{xc}|\phi^{KS}_{n\mathbf{k}}\rangle$$

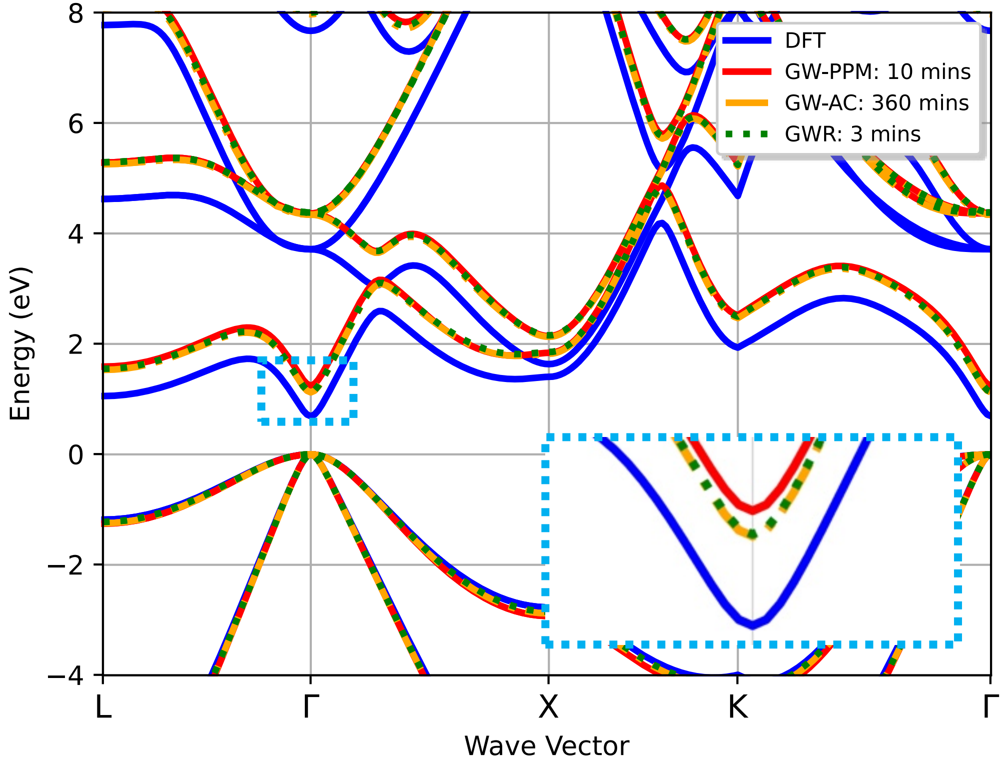
_GWR.nc

W. E. Pickett, H. Krakauer, and P. B. Allen, Phys. Rev. B 38, 2721 (1988).

Cubic and Quartic Scaling G_0W_0 (w/o SOC)

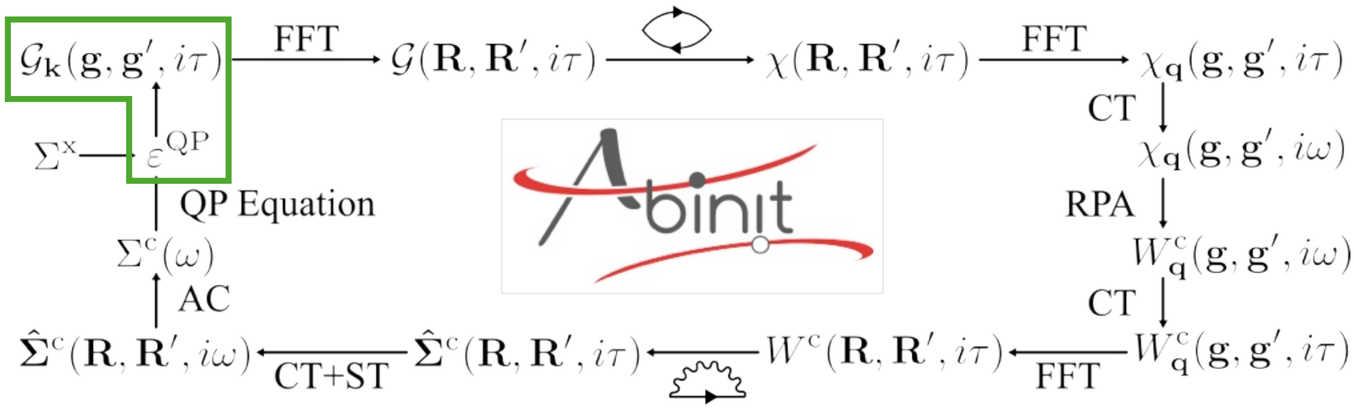


Silicon using 8 ×8 ×8 k-mesh with 256 cores



GaAs using 4 ×4 ×4 k-mesh with 512 cores

Eigenvalue self-consistent GWR (evGWR)



gwr_task:

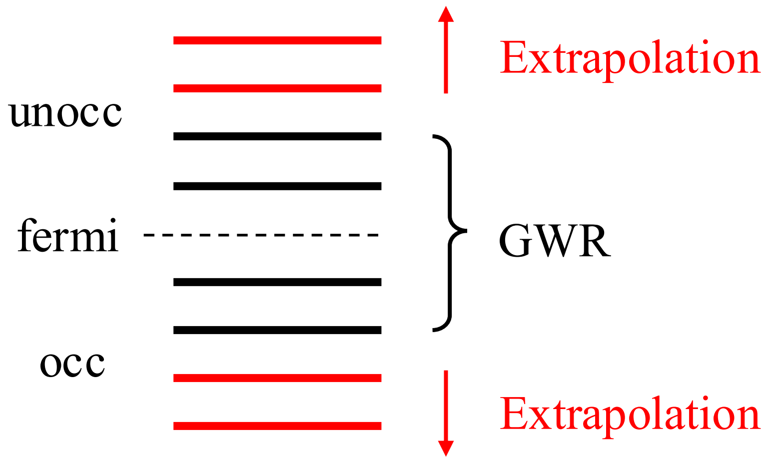
EGEW: updating G and W

EGW0: only update G

set fermi energy to zero

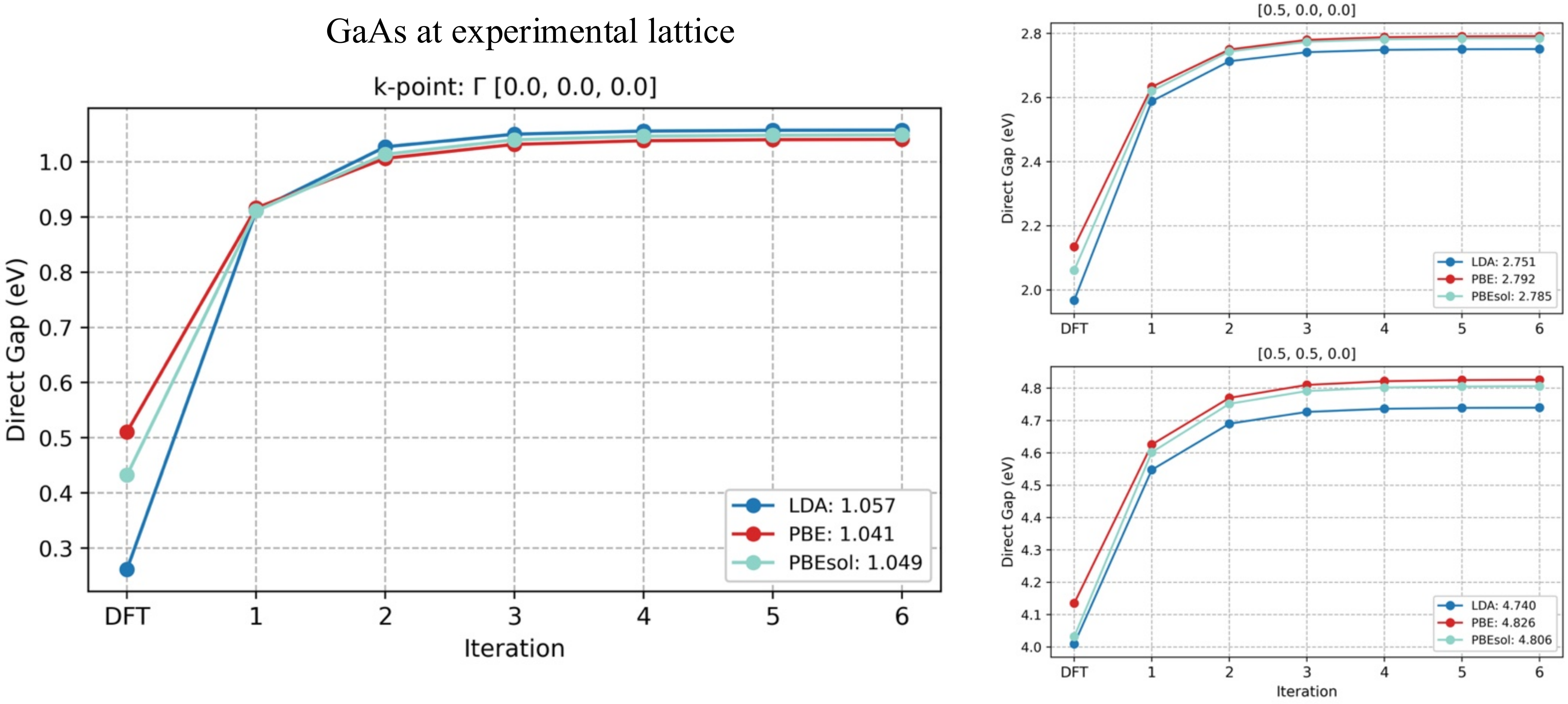
use $\varepsilon^{QP}_{n\mathbf{k}}$ to replace

$$G_{\mathbf{k}}(\mathbf{g},\mathbf{g}',i\tau) = \begin{cases} -i\sum_{n}^{\mathrm{unocc}} u_{n\mathbf{k}}(\mathbf{g})u^*_{n\mathbf{k}}(\mathbf{g}')e^{-\varepsilon_{n\mathbf{k}}\tau} & (\tau > 0) \\ i\sum_{n}^{\mathrm{occ}} u_{n\mathbf{k}}(\mathbf{g})u^*_{n\mathbf{k}}(\mathbf{g}')e^{-\varepsilon_{n\mathbf{k}}\tau} & (\tau < 0) \end{cases}$$



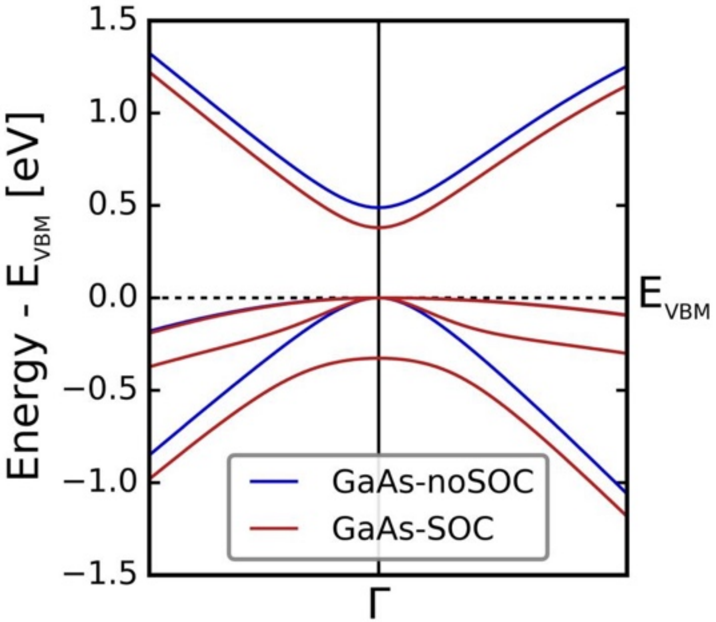
Ongoing: Dependency on starting-point

GaAs at experimental lattice



Good agreement at Γ, but not satisfactory at other high-symmetric k-points…

Cubic scaling spinor GW implementation



spin-orbit coupling (SOC): *nspinor 2*

$$\mathbf{u}_{\mathbf{k}}(\mathbf{g}) = \begin{pmatrix} u_{\mathbf{k},\uparrow}(\mathbf{g}) \\ u_{\mathbf{k},\downarrow}(\mathbf{g}) \end{pmatrix} \qquad \mathcal{G} \equiv \begin{bmatrix} G_{\uparrow\uparrow} & G_{\uparrow\downarrow} \\ G_{\downarrow\uparrow} & G_{\downarrow\downarrow} \end{bmatrix}$$

$$G_{\mathbf{k},\sigma\sigma'}(\mathbf{g},\mathbf{g}',i\tau) = \qquad (\sigma,\sigma' = \uparrow,\downarrow)$$

$$\begin{cases} -i \sum_{n}^{\text{unocc}} u_{n\mathbf{k},\sigma}(\mathbf{g}) u^*_{n\mathbf{k},\sigma'}(\mathbf{g}') e^{-\varepsilon_{n\mathbf{k}}\tau} & (\tau > 0) \\ i \sum_{n}^{\text{occ}} u_{n\mathbf{k},\sigma}(\mathbf{g}) u^*_{n\mathbf{k},\sigma'}(\mathbf{g}') e^{-\varepsilon_{n\mathbf{k}}\tau} & (\tau < 0) \end{cases}$$

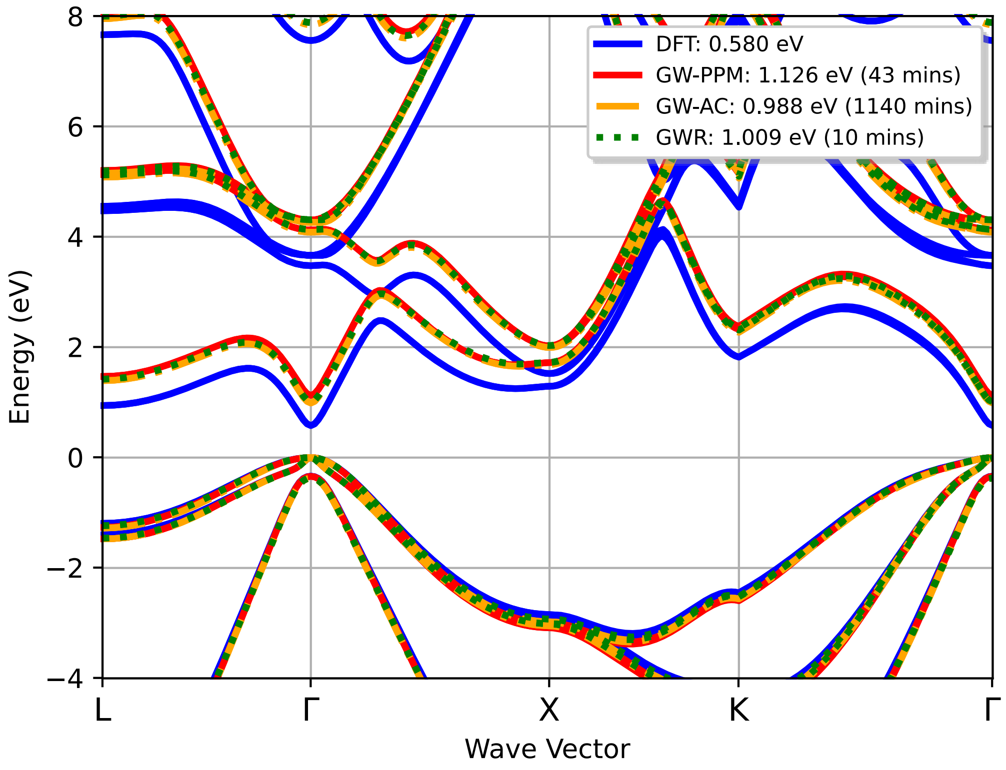
$$\chi(\mathbf{R},\mathbf{R}',i\tau) = -i \sum_{\alpha\beta} G_{\alpha\beta}(\mathbf{R},\mathbf{R}',i\tau) G_{\beta\alpha}(\mathbf{R}',\mathbf{R},-i\tau)$$

$$\hat{\Sigma}^c_{\alpha\beta}(\mathbf{R},\mathbf{R}',i\tau) = iG_{\alpha\beta}(\mathbf{R},\mathbf{R}',i\tau) W^c(\mathbf{R},\mathbf{R}',i\tau)$$

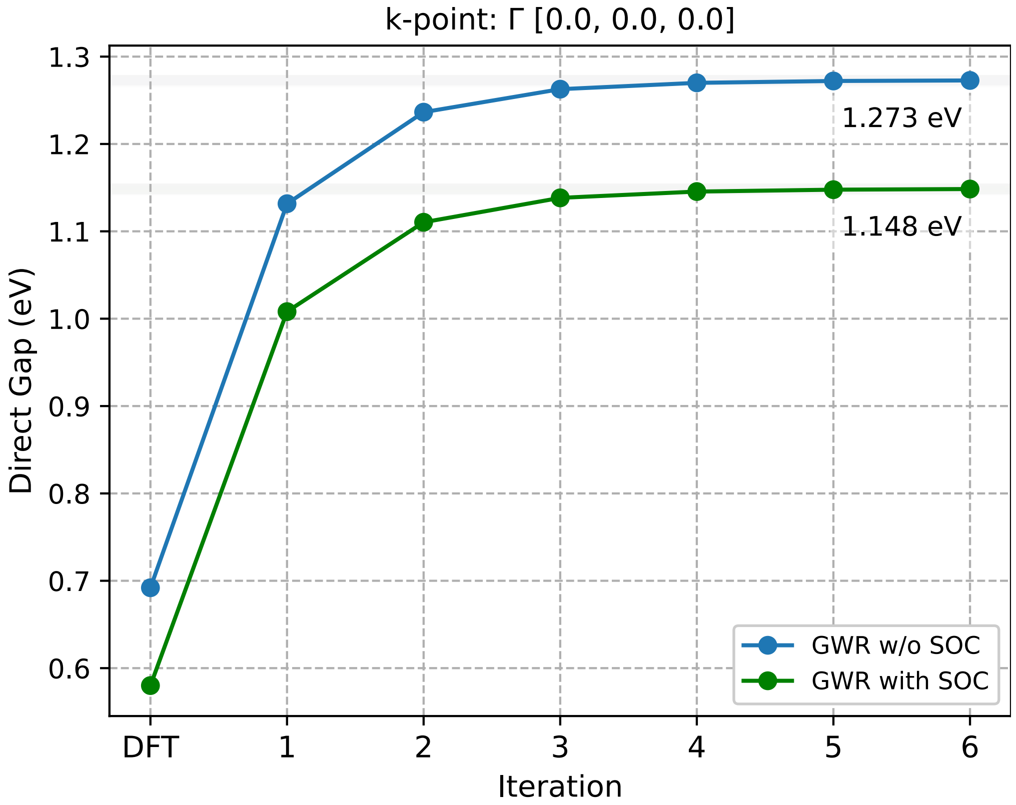
$$\Sigma^c_{mn,\mathbf{k}}(i\tau) = \sum_{\alpha\beta} \langle \psi_{m\mathbf{k},\alpha} | \hat{\Sigma}^c_{\alpha\beta}(i\tau) | \psi_{n\mathbf{k},\beta} \rangle$$

B. Maurer and A. Gulans, Spin-Orbit Coupling, http://exciting.wikidot.com/fluorine-spin-orbit-coupling (accessed Jun. 4, 2026)

Cubic scaling spinor GW application

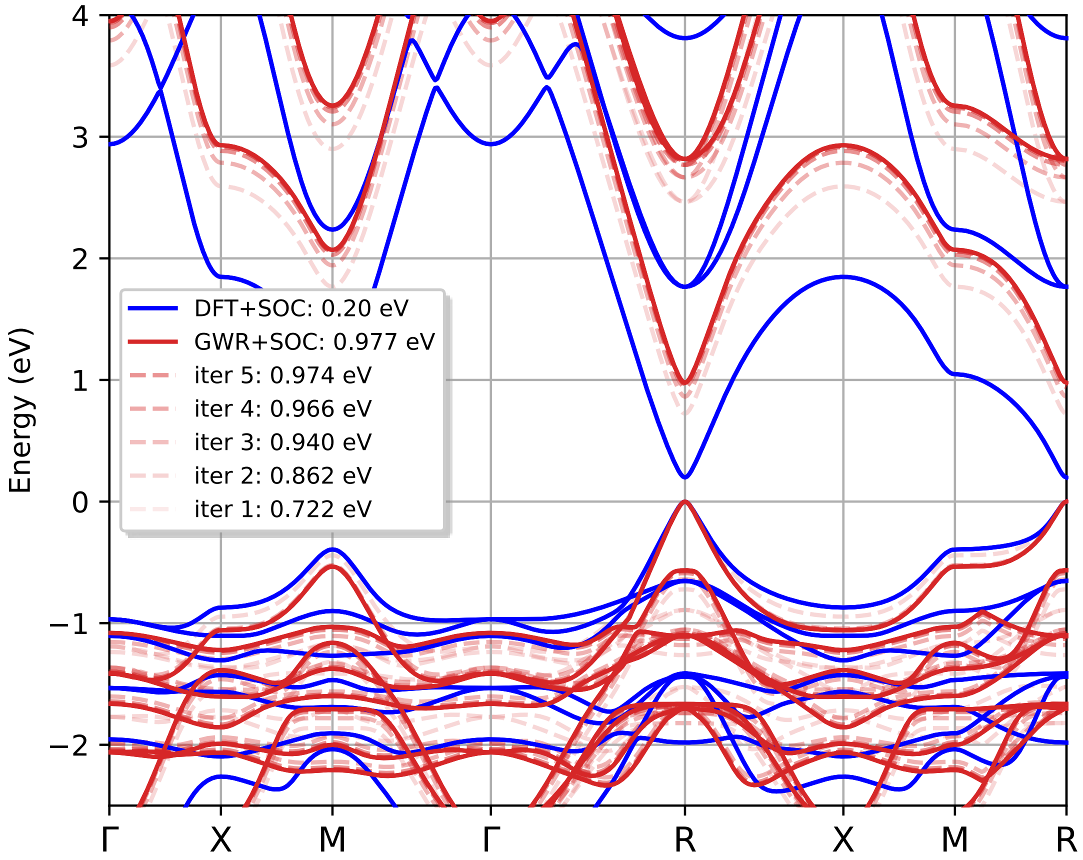


GaAs using 4 ×4 ×4 k-mesh with 512 cores (with SOC)

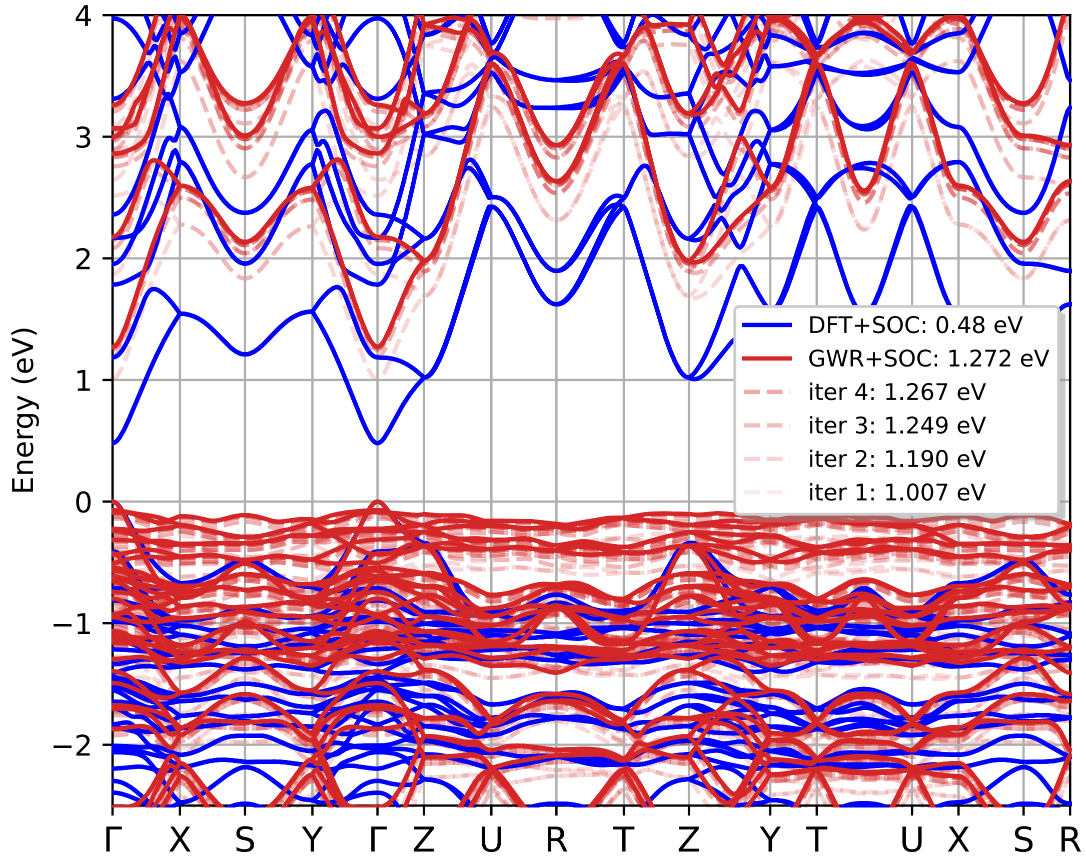


Comparison of direct band gap evGWR cycle of GaAs (with and w/o SOC)

Applications on large systems



α$CsPbI_3$ (5 atoms)



γ$CsPbI_3$ (20 atoms)

Conclusion

$$\begin{tikzcd}
& \mathcal{G}_{\mathbf{k}}(\mathbf{g},\mathbf{g}',i\tau) \arrow[r, "\mathrm{FFT}"] & \mathcal{G}(\mathbf{R},\mathbf{R}',i\tau) \arrow[r] & \chi(\mathbf{R},\mathbf{R}',i\tau) \arrow[r, "\mathrm{FFT}"] & \chi_{\mathbf{q}}(\mathbf{g},\mathbf{g}',i\tau) \arrow[d, "\mathrm{CT}"'] \\
\Sigma^{\mathrm{x}} \arrow[r] & \varepsilon^{\mathrm{QP}} \arrow[u] & & & \chi_{\mathbf{q}}(\mathbf{g},\mathbf{g}',i\omega) \arrow[d, "\mathrm{RPA}"'] \\
& \Sigma^{\mathrm{c}}(\omega) \arrow[u, "\text{QP Equation}"'] & & & W^{\mathrm{c}}_{\mathbf{q}}(\mathbf{g},\mathbf{g}',i\omega) \arrow[d, "\mathrm{CT}"'] \\
& \hat{\mathbf{\Sigma}}^{\mathrm{c}}(\mathbf{R},\mathbf{R}',i\omega) \arrow[u, "\mathrm{AC}"'] & \hat{\mathbf{\Sigma}}^{\mathrm{c}}(\mathbf{R},\mathbf{R}',i\tau) \arrow[l, "\mathrm{CT+ST}"] & W^{\mathrm{c}}(\mathbf{R},\mathbf{R}',i\tau) \arrow[l] & W^{\mathrm{c}}_{\mathbf{q}}(\mathbf{g},\mathbf{g}',i\tau) \arrow[l, "\mathrm{FFT}"]
\end{tikzcd}$$

Consistent results with conventional GW methods

Flexible and adaptable memory and parallel strategy

Supporting the spin-orbit coupling effects

Eigenvalue self-consistent calculations

UCLouvain

Acknowledgment

Supervisors:

Prof. Samuel Poncé

Dr. Matteo Giantomassi

Collaborator:

Prof. Xavier Gonze

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